FWU BSc. CSIT

--- 8th semester syllabus ---

subjects overview*:*

Eight Semester	
CSIT.421	Parallel Computing
CSIT.422	Internship
CSIT.423.2	Advanced Database Design *(Elective III)*
CSIT.424.4	Cloud Computing *(Elective IV)*
CSIT.425.2	E-Business & E-Governance *(Elective V)*

Name:
Semester:
Contact:
Address:

Course Title: Parallel Computing — **Credit: 3**
Course No: CSIT.421 — **Number of periods per week: 3+3**
Nature of the Course: Theory + Lab — **Total hours: 45+45**
Year: Fourth, Semester: Eighth
Level: B. Sc. CSIT

1. Course Introduction

In a parallel computation, multiple processors work together to solve a given problem. While parallel machines provide enormous raw computational power, it is often not easy to make effective use of all this power. This course will describe different techniques used to solve the problems, in order to develop efficient parallel algorithms for a variety of problems. We will also pay much attention to practical aspects of implementing parallel code that actually yields good performance on real parallel machines.

2. Objectives

At the end of this course, you should be able to accomplish the objectives given below.

- Describe different parallel architectures; inter-connect networks, programming models, and algorithms for common operations such as matrix-vector multiplication.
- Given a problem, develop an efficient parallel algorithm to solve it and analyze its time complexity as a function of the problem size and number of processors.
- Given a parallel algorithm, implement it using MPI, OpenMP, pthreads, or a combination of MPI and OpenMP.
- Given a parallel code, analyze its performance, determine computational bottlenecks, and optimize the performance of the code.

3. Specific Objectives and Contents

Specific Objectives	Contents
• To understand basics of parallel programming. • To explain Flynn's classification and parallel algorithm cdesign model • To design elementary parallel algorithms.	**Unit I: Parallel Programming (5)** 1.1. Introduction to parallel programming, data parallelism, functional parallelism,pipelining 1.2. Flynn's taxonomy, parallel algorithm design – task/channel model , Foster's design methodology 1.3. case studies: boundary value problem, finding the, maximum – Speedup and efficiency, Amdahl's law, Gustafson Barsis's Law, Karp-Flatt Metric, Isoefficiency metric
• To explain message passing programming model. • To understand MPI interface and use common methods provided by it • To handle timing issues in MPI programs. • To write simple programs using MPI.	**Unit II: Message Passing Programming (10)** 2.1. The message-passing model, the message-passing interface, MPI standard, basicconcepts of MPI: MPI_Init, MPI_Comm_size, MPI_Comm_rank, MPI_Send, MPI_Recv, MPI_Finalize, 2.1. Timing the MPI programs: MPI_Wtime, MPI_Wtick, collective, communication: MPI_Reduce, MPI_Barrier, MPI_Bcast,MPI_Gather, MPI_Scatter 2.2. case studies: the sieve of Eratosthenes, Floyd's algorithm, Matrix-vector multiplication

• To understand shared memory model of parallel programming and OpenMP standard. • To explain loops, critical section, function, etc in parallel programming • To write simple programs by using shared memory paradigm.	**Unit III: Shared Memory Programming (10)** 3.1. Shared-memory model, OpenMP standard, parallel for loops, parallel for pragma,private variables, critical sections 3.2. Reductions, parallel loop optimizations, general, data parallelism, functional parallelism 3.3. Case studies: the sieve of Eratosthenes, Floyd's algorithm, matrix-vector multiplication, distributed shared-memory programming, DSM primitives
• To understand basic principles of parallel algorithms • To understand principles of Monte Carlo method in algorithm design • To design parallel algorithms in specified topics.	**Unit IV: Parallel Algorithms I (10)** 4.1. Monte Carlo methods, parallel random number generators, random number distributions 4.2. Case studies: Matrix multiplication, row-wise block-stripped algorithm, Cannon's algorithm, solving linear systems, back substitution, Gaussian elimination, iterative methods, conjugate gradient method
• To design parallel algorithm for sorting data • To design searching and FFT parallel algorithms	**Unit V: Parallel Algorithms II (10)** 5.1. Sorting algorithms: quicksort, parallel quicksort, hyper quicksort, sorting by regular sampling 5.2. Fast fourier transform, combinatorial search, divide and conquer, parallel backtrack search, parallel branch and bound, parallel alpha-beta search.

Prescribed Text

- Michael J. Quinn, "Parallel Programming in C with MPI and OpenMP", TataMcGraw-Hill Publishing Company Ltd., 2003.

References

- B. Wilkinson and M. Allen, "Parallel Programming – Techniques and applications using networked workstations and parallel computers", Second Edition, Pearson Education, 2005.
- 2. M. J. Quinn, "Parallel Computing – Theory and Practice", Second Edition, Tata McGraw-Hill Publishing Company Ltd., 2002.

Course Title: Internship | **Credit: 4**
Course No: CSIT.422 | **Nature of the Course: Project**
Year: Fourth, Semester: Eight | **Level: B. Sc. CSIT**

1. Course Introduction

Practical experience in a formal work environment is a valuable aspect of a Computer Science or Computer Systems curriculum. The intent of the CS Internship program at Far Western University is to provide students with an opportunity to earn academic credit while gaining work experience at a business, government, or other institutional computer center. Students are employed on a full-time basis typically for a three to five month period. The hours, wages, and benefits associated with the job are determined by the employer prior to hiring the intern. Although tasks assigned to the student usually correspond to the student's educational background, new and exciting challenges may be encountered. Additional formal or informal training may be provided by the employer either on-site or off-site. Computer Science-related tasks, such as, network design and installation, software programming, testing, documentation, and user training would be considered as appropriate job duties for an intern. The student must be working under a mentor or expert that can provide training and guidance to the student.

2. Objectives

Students will be able to do the following:

- Apply what they have learned in the classroom.
- Learn concepts in the computing field that are difficult to teach in the classroom,such as user interaction, testing, etc.
- Experience the business and industrial environment in which a computer professional must learn to function.
- Grow professionally, emotionally, socially and intellectually.
- Sharpen their focus on career goals and course selection to reach those goals.
- Develop writing skills that are necessary in the professional world of computing.

3. Tentative Internship Report Format

The final report documents the results of the project and should be submitted within 1 week after finishing final examination. Students should use Times New Roman Font and Line spacing 1.5 while formatting their project report. Tentative project report format should be as per following outline:

Front Part

- Cover Page
- Students Declaration
- Supervisors Recommendation
- Letter of Approval
- Acknowledgement
- Abstract
- Table of Contents
- List of Figures
- List of Tables
- List of Abbreviations

Body Part

a. Organization Overview

Explain which company you interned with, where the facility was located, what the business of the company is, organization chart etc.

b. Responsibilities Handled

Explain the area you worked in and the main emphasis of your internship, Duration of Internship.

c. Discussion of Projects

Discuss in detail the areas of responsibility you had to deal with during your internship. Although this is an overview of your internship experience, include technical details about the projects you worked on. How many lines of code? What technologies, languages, tools, systems were used? Discuss the significanceof your efforts relative to the company's operations.

d. Summary and Conclusions

Summarize your work and learning experience. Explain how the internship either reinforced or changed your career goals. Discuss any new perspectivesyou obtained because of this experience. Elaborate on the benefits you realizedfrom the internship. Did you face any challenges or difficulties in yourassignments? How did you solve these issues? In what ways did you apply whatyou have learned in your graduate courses to the internship?

End Part

- References
- Bibliography
- Appendices

Note-Referencing and Citation should follow IEEE style.

4. Evaluation System

Internal Evaluation:-40% (by mentor and supervisor)

- **Proposal Defence**:-10%
  Needs to be evaluated in following basis: Organization Selection, Relevance of students intern area with CS, Presentation, Viva

- **Mid Term Evaluation**:-30%
  Students are expected to gained some experience and worked in projects. Evaluationshould be done following basis of: Efforts Made by Students, Report, Presentation, Viva

External Evaluation: - 60% (Supervisor/Mentor:-30%, External Examiner:-30%)

External evaluation should be done in the presence of external examiner and evaluation shouldbe done following basis:

1. Internship Report
2. Depth of Learning and Experience Gained
3. Presentation
4. Viva

Course Title: Advanced Database Design **Credit: 3**
Course No: CSIT.423.2 **Number of periods per week: 3+3**
Nature of the Course: Theory + Lab **Total hours: 45+45**
Year: Fourth, Semester: Eighth
Level: B. Sc. CSIT

1. Course Introduction

Advanced database design is the course that focuses on principles and algorithms of designing database management systems. This course covers concepts of file structures, indexing, query processing and optimization techniques used by database management systems. Besides this, course has given emphasis on techniques of handling transaction, concurrency, and recovery.

2. Objectives

Upon completion of the course, the student can:

→ Understand techniques and algorithm used in DBMS design
→ Demonstrate each techniques and algorithm used in DBMS design.
→ Optimize queries by creating alternative evaluation plans.
→ Develop small scale DBMS.

3. Specific Objectives and Contents

Specific Objectives	Contents
• Understand access characteristics of disks and performance parameters • Discuss role of buffer manager in performance of DBMS' • Exemplify different file organization used by database management systems	**Unit I: Storage and File Structures (8 hr)** 1.1. Physical Storage Media: Memory Hierarchy, Physical Characteristics of Disks, Performance Measures of Disks, Optimization of Disk Block Access, RAID 1.2. Storage Access, Buffer Manager, Buffer Replacement Policies 1.3. File Organization: Fixed Length Records, Variable Length Records, Organization of Records in Files, Data Dictionary Storage
• Understand need and importance of indices • Discuss different type of indices critically • Explain hashing and its applications critically	**Unit II: Indexing and Hashing (8 hr)** 2.1. Basic Concepts, Types of Indices, Factors for Evaluating Indices, 2.2. Ordered Indices, Primary Indices (Dense and Sparse), Multilevel Indices, Index update, Secondary Indices, Secondary Indices, B+ Tree Index 2.3. Static Hashing, Hash File Organization, Hash Functions, Bucket Overflow handling, Hash Indices, Dynamic Hashing, Index definition in SQL
• Understand steps of query processing • Exemplify algorithms used in performing different SQL operations • Discuss and exemplify process of evaluating SQL expressions	**Unit III: Query Processing (8 Hrs)** 3.1. Steps Involved in Query Processing, Measure of Query Cost 3.2. Select Operation: Basic Algorithms, Selection using indices, Selection involving comparisons, Implementation of Complex Selections 3.3. Join Operation: Nested Loop Join, Block Nested Loop Join, Indexed Nested Loop Join 3.4. Other Operations: Duplicate Elimination, Projection Set Operations, Outer Join, Aggregation

	3.5. Evaluation of Expressions, Materialized Evaluation, Pipelining Evaluation
• Discuss importance of optimizing queries • Exemplify size estimation of relations and its use in query optimization • Demonstrate transformation rules used in query optimization • Understand and compare cost base and heuristic query optimization	**Unit IV: Query Optimization (8 Hrs)** 4.1. Basic Concepts, Estimating Statistics of Expression Result, Catalog Information 4.2. Selection Size Estimation, Join Size Estimation, Size Estimation of other operations, Estimating Number of Distinct Values 4.3. Transformation of Relational Expressions, Equivalence Rules, Examples of Transformations 4.4. Cost Based Query Optimization, Heuristic Query Optimization, Optimization of Nested Queries
• Understand basic concept of transaction and interleaved processing • Discuss need of serailizable schedules • Exemplify serializability test procedure	**Unit V: Transaction Management (4 Hrs)** 5.1. Basic Concepts, ACID Properties, Transaction States, Concurrent Execution 5.2. Schedules, Types of Schedule on the Basis of Serializability, Testing Conflict Serializability, Types of Schedule on the Basis of Recoverability 5.3. Commit and Rollback
• Understand need of concurrency control • Discuss different protocols used in controlling concurrency and exemplify each of them • Exemplify techniques of handling deadlocks	**Unit VI: Concurrency Control(5 Hrs)** 6.1. Lock Based Protocols, Timestamp Based Protocols, Thomas write Rule 6.2. Validation Based Protocols, Granularity, Multiversion Protocols 6.3. Deadlock Prevention (wound-wait and wait-die), Deadlock Detection, Recovery from Deadlocks
• Discuss need of recovery techniques • Exemplify log based recovery schemes • Explain shadow paging technique of recovery	**Unit VII: Recovery System(4 Hrs)** 6.4. Types of Failures, Recovery Schemes, Log File, Write Ahead Logging 6.5. Log Based Recovery Techniques (undo/redo, no-undo/redo, undo/no-redo), Check pointing, Shadow Paging 6.6. Recovery in concurrency

Prescribed Text

- **Database System Concepts,** by Abraham Silberschatz,, Henary Korth, S. Sudarshan,McGraw-Hill Education, Sixth Edition, 2010
- Raghu Ramakrishnan, and Johannes Gehrke, Database Management Systems, 3rd Edition ,McGraw-Hill, 2007
- Ramez Elmasri and Shamkant B. Navathe, Fundamentals of Database Systems, 6thEdition, Pearson Addison Wesley; 2010

Course Title: Cloud Computing **Credit: 3**
Course No: CSIT.424.4 **Number of periods per week: 3+3**
Nature of the Course: Theory + Lab **Total hours: 45+45**
Year: Fourth, Semester: Eighth
Level: B. Sc. CSIT

1. Course Introduction

The course introduces the ideas and techniques underlying the principles of cloud computing. This course covers a series of current cloud computing technologies, including technologies for Infrastructure as a Service, Platform as a Service, and Software as a Service. This course is designed to introduce the concepts of Cloud Computing as a new computing paradigm. The students will have an opportunity to explore the Cloud Computing various terminology, principles and applications. The course will expose students to different views of understanding the Cloud Computing such as theoretical, technical and commercial aspects.

2. Objectives

The main objective of this course is:

- To provide students with the fundamentals and essentials of Cloud Computing.
- To provide students a sound foundation of the Cloud computing so that they are able to start using and adopting Cloud Computing services and tools in their real life scenarios.
- To provide the knowledge about the SOA, cloud security and cloud disaster management

3. Specific Objectives and Contents

Specific Objectives	Contents
• Understand basics cloud framework • Understand concepts of cloudcomputing • Understand the features of cloud computing • Understand the cloud deployment models	**Unit I: Introduction (9 Hrs)** 1.1. Cloud, Cloud computing, Components of cloud computing, Characteristic features of cloud computing, 1.2. Evolution of cloud computing, Challenges for the cloudcomputing, Benefits of cloud computing, 1.3. Grid computing, Cloud Computing vs Grid Computing,Distributed Computing in Grid and Cloud, 1.4. Cloud deployment models: Public, Private, Hybrid,Community, 1.5. Cloud Service Models: IaaS, PaaS, SaaS, 1.6. Challenges for cloud computing, Legal issues in cloudcomputing.
• Understand concepts of virtualization and it approaches • Explore concepts of virtualization in cloud environment	**Unit II: Virtualization (5 Hrs)** 2.1. Basic Concepts of virtualization, Hardware virtualization, Server virtualization, Storage virtualization, Data Centre virtualization OS virtualization, Para virtualization, 2.2. Role of virtualization in enabling cloud services, Cloud computing as a virtualized service.

• Understand the cloud migrationand its need • Explore the cloud migrationmodel • Determine the risks during cloud migration	**Unit III: Cloud Migration(4 Hrs)** 3.1. Cloud Migration and its types, Need for CloudMigration, 3.2. Model of Migration into a cloud, 3.3. Migration risks in Cloud and Mitigation.
• Understand various cloudservice models • Understand and analyze variousaspects of the cloud service models • Explore the real world cloud services	**Unit IV: Cloud Service Models (15 Hrs)** 4.1. Infrastructure-as-a-Service (IaaS), 4.2. Platform-as-a-Service (PaaS), Key Characteristics ofPaaS, 4.3. Software-as-a-Service (SaaS): SaaS ImplementationIssues, Key Characteristics of SaaS, Benefits of the SaaS Model, 4.4. Communication-as-a-Service (CaaS): Advantages of CaaS, 4.5. Monitoring-as-a-Service (MaaS), 4.6. Jericho Cloud Cube Model, 4.7. Amazon's Web Services, 4.8. Cloud Computing from the Google Perspective, 4.9. Window Azure and Online Services.
• Understand Service OrientedArchitecture (SOA) • Explore significance of SOA in Cloud Computing	**Unit V: SOA and Cloud (4 Hrs)** 5.1. Service Oriented Architectures (SOA), 5.2. Combining the cloud and SOA 5.3. Characterizing SOA, 5.4. Importance of SOA to cloud computing
• Understand security in cloud • Understand risk assessment incloud • Explore various intrusion detection mechanisms in cloud environment. • Understand how to handlecloud disasters and how to mitigate the disaster	**Unit VI: Cloud Security (8 Hrs)** 6.1. Cloud Security Challenges, Dimensions of Cloud Security: Security & Privacy, Compliance, and Legal or Contractual Issues, 6.2. Risk Management, Security Monitoring, Incident Response Planning, Security Architecture Design, Vulnerability Assessment, Data and Application Security, Virtual Machine Security, 6.3. Handling Disasters in Cloud, Disaster Recovery, Disaster Recovery Planning, Disaster Management.

References

- Dan C. Marinescu, *Cloud Computing: Theory and Practice*, Morgan Kaufmann.
- Rajkumar Buyya, The University of Melbourne and Manjrasoft Pty Ltd., Australia, JamesBroberg, The University of Melbourne, Australia Andrzej Goscinski, Deakin University, Australia, C*loud Computing Principles and Paradigm,* John Wiley and Sons Inc. Publication.
- John W. Rittinghouse and James F. Ransome, *Cloud Computing: Implementation Management and Security*,
- George Reese, *Cloud Application architecture*, O'Reilly Media Inc.
- Judith Hurwitz, Robin Bloor, Marcia Kaufman, Fern Halper, *Cloud Computing forDummies,* Wiley Publishing Inc.
- Borko Furht, Armando Escalante, *Handbook of cloud computing*, Springer, 2010
- David S. Linthicum, *Cloud Computing and SOA Convergence in your Enterprise*, a step by step guide, Addison Wesley

Course Title: E-business & E-Governance **Credit: 3**
Course No: CSIT.425.2 **Number of periods per week: 3+3**
Nature of the Course: Theory + Lab **Total hours: 45+45**
Year: Fourth, Semester: Eighth
Level: B. Sc. CSIT

1. Course Introduction

This course is aimed to understanding the concept of e-Governance to better delivery of government services to citizens, improved interactions with business and industry, citizen empowerment through access to information, efficient government management and resulting benefits can be less corruption, increased transparency, greater convenience, revenue growth and cost reductions.It cover the concept of e-Governance, different model of e-Governances and maturity levels, infrastructure and readiness for e-governance, data ware house and data mining for e-government services, initiatives in Nepal and recent trends of e-Government issues. Student will be analysis the major e-governance case study of Nepal and best case studies of aboard.

2. Objectives

After completion of course, Students will be able to:

- Understands the basic principle of e-Governance and importance of digital world.
- Analysed the different model of digital governance and its maturity levels.
- Define the e-Readiness to successful implementation of e-Governance and analyse current situation of Nepal.
- Determine the importance of data mining and data warehouse and open data in e-Governance.
- Analyse the situation of e-Governance in Nepal.
- Analyse the case study about different e-Government Projects.

3. Specific Objectives and Contents

Specific Objectives	Contents
• Define e-Governance and importance • Explore changing nature of e-Governance services • List out the present global trends of e-Governance • Compare government and governance	**Unit I: Concept of e-Governance(10Hrs.)** 1.1. Definition of e-Governance 1.2. Importance of e-Governance 1.3. Evolution of e-Governance: Its scope and Contents 1.4. Present Global Trends of Growth in e-Governance 1.5. Differentiate Between e-Government and e-Governance
• Analyze the different digital model of e-Governance • List of level of maturity model and its parameters. • Justify e-Governance toward good governance.	**Unit II: e-Governance Models(15 Hrs.)** 2.1. Model of Digital Governance 2.1.1 Broadcasting Dissemination Model 2.1.2 Critical Flow Model 2.1.3 Comparative Analysis Model 2.1.4 Mobilization and Lobbying Model 2.1.5 Interactive-Service Model/ Government-to-Citizen-to-Government (G2CG2G)Model 2.2. Evolution of e-Governance and Maturity Models 2.3. Characteristics of Maturity Model 2.4. Key Focus Area 2.5. Toward good governance through e-Governance Model

• Identify the e-Readiness parameters to success of e-government. • Analyzed the situation of e- Governance readiness in Nepal	**Unit III: e-Governance Infrastructure, Stage in Evolution andStrategic for Success (15)** 3.1. e-Readiness 3.1.1 Data System Infrastructure 3.1.2 Legal Infrastructure Preparedness 3.1.3 Institutional Infrastructure Preparedness 3.1.4 Human Infrastructure Preparedness 3.1.5 Technical Infrastructure Preparedness 3.2. Evolutionary Stage in e-Governance
• Describe the importance of data warehouse and mining in e-Government services. • Explore the area of data warehouse and data mining on governance services.	**Unit IV: Application of Data Warehouse and Data Mining in Government (5Hrs.)** 4.1. National Data Warehouses 4.2. Area for Data Warehouse and Data Mining 4.3. Big data in e-Governance
• Understand the open standards and GA of Nepal • Review the status of government data center in Nepal • Describe the e-Government related Act and policies of government of Nepal.	**Unit V: e-Governance of Nepal (10Hrs.)** 5.1. Evolution of e-Governance in Nepal 5.2. Government Enterprises Architecture(GEA) 5.3. E-Government Master plan 5.4. GIDC and Data Centre 5.5. Electronic Traction Act 2063 5.6. Information Communication Technology Policy 2072 5.7. Digital signature
• Understand recent trends in e-Governance • Describe e-Democracy • Describe internet governance • Understands the web standardto e-Governance.	**Unit VI: Recent Trends in e-Governances (15Hrs.)** 6.1. e-Government 2.0: Next Generation Governance 6.2. e-Democracy 2.0 6.3. Open Data: Definition, Principle, uses 6.4. Mobile Governance 6.5. Open Standards for Web Presence 6.6. Government Cloud Services and Open Sources
• Analyze the case study of case study of Nepal • Analyzed selected case study of successful e-Government project. • Create the report of case study	**Unit VII: Case Study (20Hrs.)** 7.1. ICT Development Project ADB in Nepal 7.2. National ID in Nepal 7.3. Government Electronic Procurement System of Nepal(GEPSON) 7.4. IT park Kavre, Banepa 7.5. e-Village/Tele Centre in Nepal 7.6. Smart City in Nepal 7.7. Digital India Project in India

Case Study

Student should analyses the case study of e-Governance practices. Students are recommended to visit to data center, e-Village and Tele-center among countries. The case study should be practiced for one case study per week.It is highly recommended that prepared case study report and presentation on group which is found in study period. A group of four or five students can work together.

Prescribed Texts

- Prabhu, C. S. R. (2012). *E-governance: concepts and case studies*. New Delhi: Prentice-Hall ofIndia.

References

- Srinivas Raj, B. (2008). *E-governance techniques: Indian and global experiences*. New Delhi,India: New Century Publications.
- Bhatnagar, S. C. (2009). *Unlocking e-government potential: concepts, cases and practical insights*. New Delhi, India : Thousand Oaks, Calif: SAGE.
- Agarwal, A. (Ed.). (2007). *E-Governance: case studies*. Hyderabad: Universities Press.UN E-Government Survey 2016: http://www.unpan.org/
- Electronic Transaction Act 2063: http://www.lawcommission.gov.np/
- ICT Policy 2072: http://moic.gov.np/np/
- E-Villages and Tele centers: http://doit.gov.np/
- GIDC: http://nitc.gov.np/

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